

Forecasting Annual U.S. Honey Bee Colony Numbers Using Time Series Data

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1 Problem Statement

North America is home to nearly 4,500 species of bees [1]. Most species are solitary, but there are 49 known species of primitive eusocial bumble bees in the United States. 41 of these species are also found in Canada, and an additional 11 species are found in Mexico. In 1622, the highly eusocial western honey bee (“*Apis mellifera*”) was introduced to North America from Europe and Africa [2], and it is the only species of honey bee in North America.

Honey bees and other plant pollinators are vital to the health of agricultural and plant ecosystems. In fact, honey bees pollinate many of the fruits, vegetables, and nuts the human population grows and consumes. 75% to 80% of the world’s crops depend on pollinators like the honey bee [3], and the need for honey bee pollination services has increased as the demand for insect-pollinated food has risen. Unfortunately, honey bees in the United States have a high mortality rate. “...over the last decade, the U.S. annual honey bee colony mortality rate has fluctuated between 30-45%, with some beekeepers losing half or more of their colonies every year” [4]. Nonetheless, colony loss rates remain separate from honey bee population decline. The honey bee population (i.e., the overall number of colonies) has remained steady over the years, but the beekeeping and agricultural industry do face challenges [5]. There is still notable concern for the overall health of honey bee colonies, which has led to the establishment of nonprofit organizations, such as “The Bee Informed Partnership”, to help improve the health and long-term sustainability of honey bees.

In the United States, honey bees have such a high mortality rate due to a complex suite of factors. “Several critical factors threaten honey bees including poor nutrition due to land use change, pesticide exposure, and introduced pests and diseases” [4]. Varroa mites, which were introduced into the United States in the 1980s, are one of the major external parasitic threats that continue to negatively impact US honey bee colonies. Elevated pathogens and pesticides also have a substantial influence on honey bees, with a wide array of pesticides being detected in wax and pollen [4]. These external threats then make honey bee colonies more susceptible to serious diseases, like “Colony Collapse Disorder” (CCD), which is a phenomenon that occurs when thriving colonies empty within a few weeks because many worker bees disappear [6].

Moreover, climate change continues to have intensifying weight on honey bee mortality, especially due to extreme temperatures. To elaborate, honey bees are active when it is 60.8 degrees Fahrenheit, but they stop flying when the temperature falls below 50 degrees. If the temperature drops below 28 degrees, then they die within one hour [7]. In a survey performed from 2014 to 2019, the mortality rate for bees was 53.5% during the winter [8]. While honey bees can survive in both high and low temperatures, they will still die if there are prolonged instances of substantially high or low temperatures because they suffer severe water loss and weakened muscles to fly [7].

Finally, air quality can damage honey bee colonies. A 2020 study performed in India demonstrated that the number of bees visiting flowers reduced significantly in areas with elevated levels of pollution, while the opposite was true in areas where pollution was lower [9]. The study identified high and low pollution areas by investigating the levels of particulate matter (PM), consisting of sulfate, nitrates, ammonia, sodium chloride, black carbon, mineral dust, and water, in the air [9]. Note that elevated levels of PM_{2.5} means that the PM is small enough to enter the systems of humans and animals [10]. Other environmental exposures such as ozone, nitrogen, and sulfur dioxide can also have serious outcomes on honey bee health, and elevated levels of pollutants can cause bees to die.

Consequently, such highly volatile dynamics in honey bee mortality can lead to unpredictable outcomes for our country's ecosystems and agricultural industry. Colony loss directly affects the honey bee population, and a potential decrease in the number of honey bee colonies could negatively impact crop production yields. While beekeepers can help to create more colonies, replacing colonies still comes at a great cost, and it should not be underappreciated [4]. Over the last 20 years, annual loss rates remain high, so beekeepers are consistently under pressure to offset these losses by creating new colonies [11]. Instead of reactionary decision making based on constant fluctuations in the honey bee population, it could help beekeepers, farmers, etc. to have insight into how many colonies they can expect to have in the future. This would enable the appropriate policy makers to improve assessments of local and regional habitat suitability and ultimately make more informed decisions. If these groups had advanced knowledge of substantial colony loss in the future, then perhaps they could take steps in the present to mitigate some of the problems through targeted initiatives. Therefore, the goal of this project is to utilize past data to help build a machine learning model that forecasts annual honey bee colony numbers for different states in the U.S.

2 Literature Survey

As explained, the topic of honey bee population loss is an important one. Most of the research being done is focused on analyzing data collected through surveys. The "Bee Informed Partnership" created the "Colony Loss and Management Survey," which monitors managed honey bee colony loss rates and identifies risk factors and protective measures [12]. Furthermore, the "National Honey Bee Survey" is a comprehensive examination of colony health throughout the United States, and it helps document honey bee diseases, pests, and pathogens [13]. The data collected through these surveys is valuable to both beekeepers and researchers.

For instance, a review published in *EcoHealth* [14] utilized data collected through surveys from individual beekeepers to study the drivers of decline in bee colonies. The study was conducted by identifying socioeconomic, political, and environmental factors affecting the bee industry, and various models were run to analyze bee colony numbers. It was found that how the bees are managed, along with pathogens and pesticides, were significant contributors to the decline of the bee colonies. The approach used in the study can be improved by adding more datasets and evaluating data across the entire United States to get a more extensive view of the factors impacting honey bee colonies, such as weather and air quality, which our project intends to do. Most importantly, our primary goal is to create a machine learning model to forecast future honey bee colony numbers.

However, there are other studies that have taken a more theoretical approach. Chen et al. [15] formulated theoretical mathematical equations to study the impact on the number of honeybee colonies, and they considered several factors as variables, such as diseases, pesticides, and food availability. Though, the equations were only based on theory, and they were not validated for practical use. The authors did state that the use of machine learning to validate the theoretical models was necessary since their approach lacked the use of real data. Our project will use machine learning modeling to validate real historical time series data,

and we'll consider various factors when building our machine learning models.

Regardless, there are still only a few studies related to forecasting honey bee activity or colony numbers. Ngo et al. [16] used temporal convolutional neural networks to forecast the population daily loss rate of honey bee colonies, and they determined warning levels based on the predictions. Yet, their project was narrowly focused on individual honey bee colonies since they collected experimental data from a honey bee farm located in Taiwan to help construct their algorithm. Moreover, Rumkee et al. [17] simulated the effects of pesticides on different honey bee life stages using the BEEHAVE model. "The BEEHAVE model simulates the development of a honey bee colony and its foraging of nectar and pollen in different landscapes" [18]. Matthias et al. [19] was another study that utilized the BEEHAVE system. The authors found the increase in failure of bee colonies were attributed to pathogens, forage availability, and potentially pesticides. To sum up, all these initiatives focused on how individual honey bee colonies responded to external stressors, specifically pesticide exposure and pathogens.

Our study, on the other hand, will have a broader focus than the research projects outlined in the previous paragraph since we will try to predict the annual honey bee population for individual states, as well as the aggregated numbers of these states. In addition, we will try to consider several factors that are correlated with increased honey bee mortality, including pesticides, diseases, pests, temperature, precipitation, and air quality. The aim of this project is to account for the factors that have a potential impact on honey bees to make more informed predictions. Finally, we will experiment with different machine learning models to see which approach produces the best forecasts of future honey bee colony numbers.

3 Data Sources

For our project, we produced two separate data sources to experiment with different kinds of methodologies because we wanted to examine which ones produced the best predictive results.

3.1 Dataset A

To easily distinguish between the two datasets we utilized, we referred to the dataset described in this subsection as "Dataset A." The primary segment of the data was retrieved from the National Agricultural Statistics Service (NASS), which is part of the United States Department of Agriculture (USDA) [20]. It contained annual honey bee colony numbers, which we treated as our response variable, for various states from 1987 to 2021. Important to note, this was the most complete and expansive dataset on honey bee populations that could be obtained from the NASS. The NASS creates its datasets through a wide-ranging Census conducted every 5 years, or through more constrained quarterly or annual surveys. Our group decided to focus on the data collected from the surveys.

Unfortunately, due to the limitations of these surveys, there were 11 states that did not have observations for every year. Instead, some of these states had their honey bee numbers grouped together in a separate series labelled "other states." Since the set of states that made up "other states" were widely dissimilar and ever changing, we decided to remove this series from the dataset entirely. Furthermore, any states that did not have continuous observations for the entire time horizon were also dropped. Therefore, the final version of Dataset A was composed of separate time series of annual honey bee colony numbers for 39 states from 1987 to 2021, and we assumed these 39 states were a fair representation of the United States as a whole.

With this setup, Dataset A was often treated as a hierarchical time series. The summation of the honey bee population in a given year across the 39 states represented the "top" of the hierarchy, or the most aggregate level of the data. So, the t -th observation of the aggregated time series was denoted by y_t for $t = 1987, \dots, 2021$. Obviously, this top-level series was also disaggregated into the 39 separate time series,

where $y_{j,t}$ denoted the t -th observation corresponding to the j -th series for $j = 1, \dots, 39$. For any time t , the observations for all the j time series summed to the aggregated time series, i.e.,

$$y_t = y_{1,t} + \dots + y_{39,t}$$

Moreover, we generated annual data related to climate and air quality for the 39 states. These features were frequently combined with the annual honey bee colony numbers to create multivariate time series for each state and help improve the predictive power of the models employing Dataset A. First, the climate data was retrieved from the National Centers for Environmental Information (NCEI) [21]. The dataset consisted of monthly weather measurements at the state level from 1895 to the present. For our project, the maximum temperature, minimum temperature, and precipitation measures were retained. During the preprocessing, the monthly data was rolled up to the annual level for each state by taking the monthly maximum and minimum temperatures, respectively, over the 12-month readings, while the annual precipitation was calculated by adding the 12-month readings. Next, the air quality data was obtained from the United States Environmental Protection Agency (EPA) [22]. The annual summary data per county was extracted for the years of 1987 to 2021, and the data was preprocessed to merge all the various data files sets into one dataset. For each feature column on air quality, the data was grouped into states and counties, and the aggregate mean for the 39 states was calculated using the appropriate county values.

3.2 Dataset B

Moving on, we referred to the dataset described in this subsection as “Dataset B.” This dataset contained annual honey bee data for 45 states and “other states” (an aggregation of data from Alaska, Delaware, Nevada, New Hampshire, and Rhode Island) from 2015 to 2021. Similar to the other dataset, much of the data in this dataset was obtained from the NASS quarterly survey data [20]. Though, Dataset B did not only include annual data on honey bee populations. It also contained data on various features that typically impact honey bee mortality, such as the percentage of honey bee colonies affected by renovation, diseases, pests, pesticides, and “other causes”. The quarterly data for each state was aggregated to the annual level by taking the average of the four quarters for each feature. We also obtained annual data on the percentage of beekeepers and honey bee colonies exclusive to each state, along with a weighted average of total colony loss, from the “Bee Informed Partnership” [23]. All these additional features were been combined with the annual data on air quality and climate to produce a large set of independent variables.

Now, the main goal of our project was to generate a strong predictive model to forecast annual honey bee colony numbers in the future. To accomplish this objective, we wanted to utilize both datasets to generate and test several models, and then select the model with the best predictive performance. Yet, in order to properly compare models built using either of the two datasets, the set of states in both needed to match. So, we eliminated data corresponding to “other states” and the 6 additional states found in Dataset B (i.e., Connecticut, Maryland, Massachusetts, New Mexico, Oklahoma, and South Carolina).

Finally, the structure of Dataset B was more in line with longitudinal or panel data. As mentioned, it not only had data for our response variable, but it also consisted of data for various independent variables. The dataset contained a collection of various quantities obtained across multiple observational units, in this case the 39 individual states, that were assembled over even intervals in time and ordered chronologically. So, for each observational unit i in Dataset B, we had a time series $\{(x_t, y_t) : t = 2015, \dots, 2021\}$ with a total of $T = 7$ years where x_t was a $n \times 1$ feature vector and y_t was the response variable, and we had a total of $T = 7$ years. Therefore, we could write the data for all our $m = 39$ observational units compactly in matrix

form $\{(\mathbf{X}^i, \mathbf{y}^i) : i = 1, \dots, 39\}$ where $\mathbf{X}^i$ was a $T \times n$, written as

$$\mathbf{X}^i = \begin{bmatrix} (x_1^i)^T \\ (x_2^i)^T \\ \vdots \\ (x_T^i)^T \end{bmatrix}$$

x_t^i was our standard $n \times 1$ feature vector, and $\mathbf{y}^i$ was a $T \times 1$ vector of response variables.

3.3 Similarities and Differences

Now, for both of the datasets, we collected data for 39 U.S. states, or "units", over fixed periods of time. In addition, the two datasets were "fixed and balanced" since each state was tracked over the same number of time periods, and the same set of states were tracked in each time period. For each unit, we included a set of explanatory variables, which all changed across time periods. We wanted to make use of these measured features to help forecast the response variable, which was the total number of bee colonies.

Yet, after aligning the set of states, there were two major differences between Datasets A and B: Dataset A had a larger time horizon than Dataset B, while Dataset B contained more explanatory variables. To elaborate, the time frame of Dataset A consisted of 35 years, while the time frame of Dataset B was only 7 years. Dataset B, on the other hand, contained features that were not part of Dataset A. As mentioned previously, these additional features should have some impact on honey bee colony numbers.

Above all, our goal was to choose and fit models that were appropriate for whichever dataset was deployed. In fact, the differences in structure between the two datasets influenced the methodologies that we implemented. The number of time periods and "units" in Dataset A were almost equal, and Dataset A had a longer time frame. So, we experimented with time series methods that performed better under these circumstances by fitting them on Dataset A. With Dataset B, there was a small number of time periods, and the number of "units" was much larger. This meant that regression models more suited for panel datasets were fitted using Dataset B.

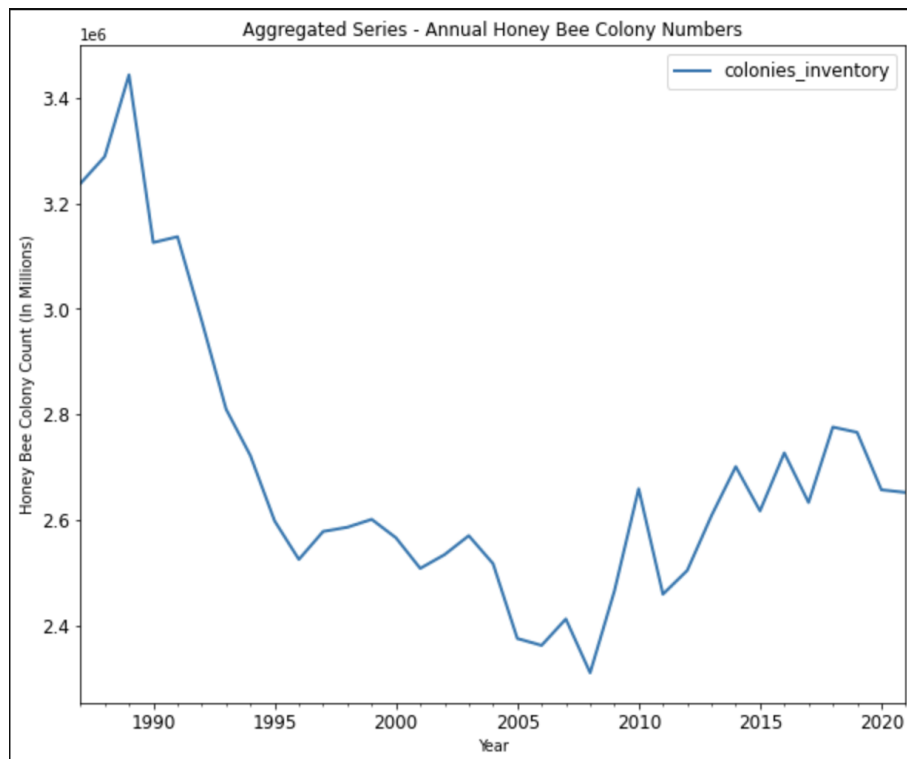
4 Methodologies

Prior to model fitting and evaluation, our group performed exploratory data analysis on both datasets to help inform our decisions in regards to data preparation, variable selection, and model building. Then, using both datasets, we experimented with several models to see which one produced the best predictive results. In the subsequent subsections, we explain some of our notable findings about the datasets, and we outline the different modeling approaches we tested.

4.1 Exploratory Data Analysis - Dataset A

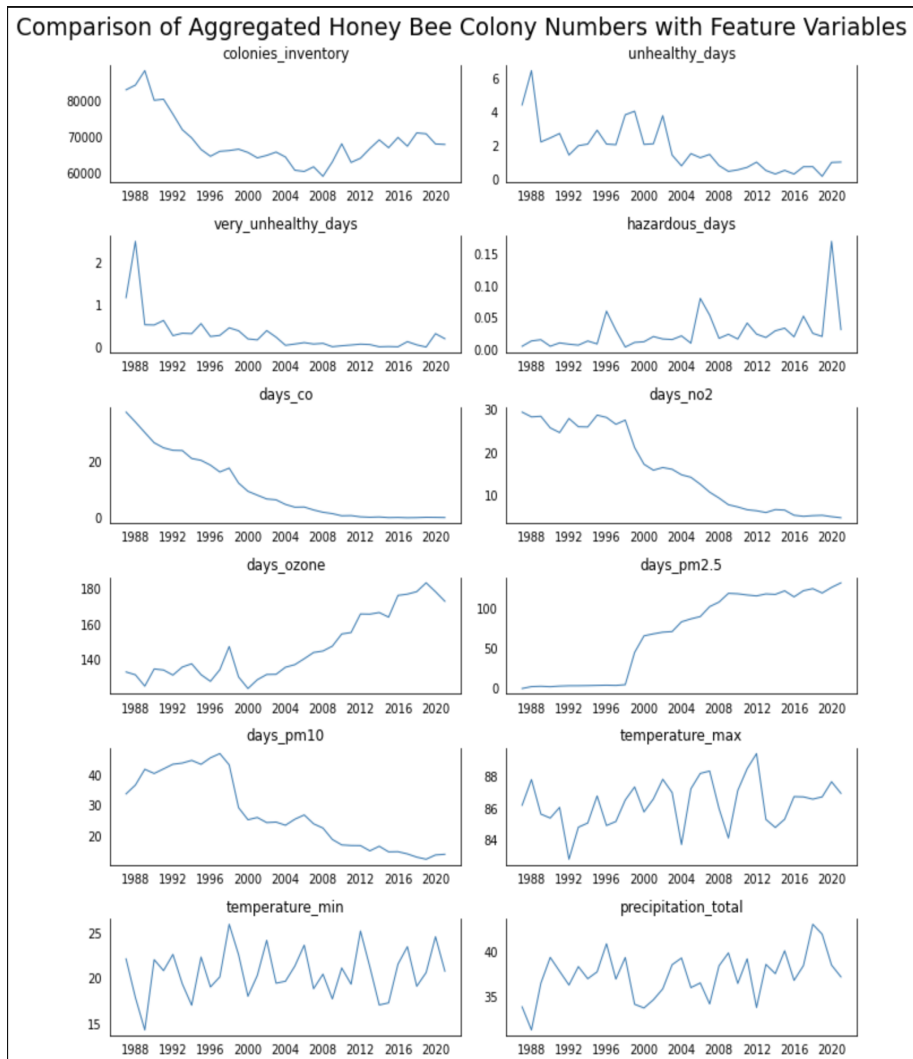
4.1.1 Visualizations

With a large number of states in our dataset, we wanted to prioritize our focus. So, we decided to analyze the overall trend of the annual honey bee colony numbers of the aggregated time series ($y_t = y_{1,t} + \dots + y_{39,t}$), as opposed to the time series of each individual state.

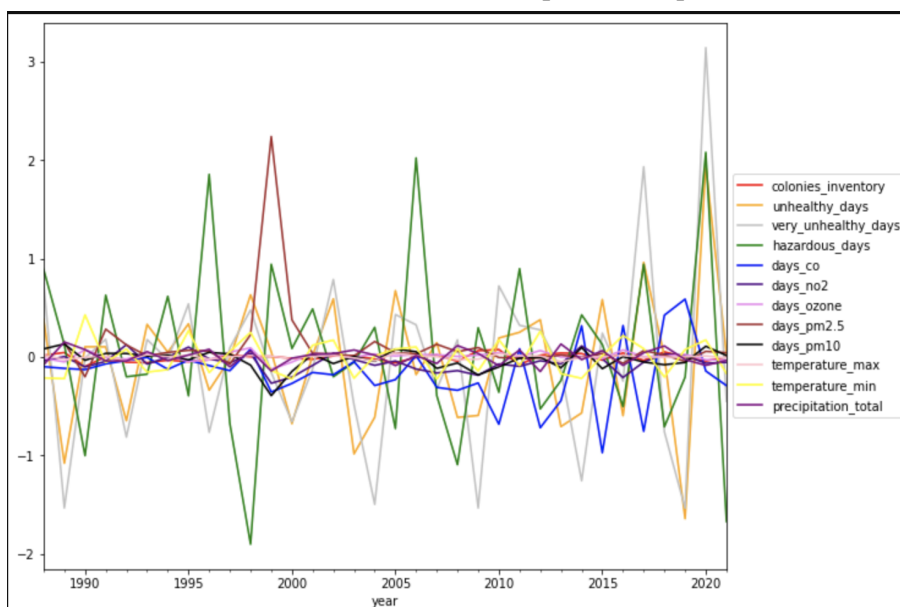


There was a substantial decrease in annual honey bee colony numbers for the aggregated time series from 1987 to 2008, but since 2008, the total numbers have slowly started to rise and recover.

In addition, we wanted to compare the aggregated time series of the response variable with the aggregated time series of the several climate and air quality features. As stated above, each of the 39 states had distinct time series for all of the different factors. Thus, to generate the aggregated time series for a given feature, we simply took the average across the 39 states for each year. Here is a raw comparison between the aggregated time series of the response variable and the aggregated time series of the various features.



We also formulated a normalized view to compare the response variable with the feature variable.



From the visualizations above, the most important features affecting the aggregated time series of annual honey bee colony numbers ($y_t = y_{1,t} + \dots + y_{39,t}$) seemed to be “hazardous_days”, “days_pm2.5”, and “days_ozone”. These variables represented the average number of days in a given year in which air quality was deemed “hazardous” and there were elevated levels of PM2.5 and ozone, respectively. All of three of these variables should negatively impact honey bee mortality rates, and we can see that while these variables had a positive trend, the honey bee colony numbers had an overall negative trend.

4.1.2 Statistical Tests

Still, we felt that it was important to conduct a few statistical tests on the different time series before making any final decisions about feature selection. It is important to note that all of the statistical tests outlined below were conducted on the aggregated time series of the response variable ($y_t = y_{1,t} + \dots + y_{39,t}$) and the aggregated time series of the various climate and air quality features.

First, we performed the Granger causality test for different sets of variables. When constructing models related to time series, like Vector Autoregression (VAR), it was important that each of the time series in the system influenced each other. With the Granger causality test, we tried to determine which time series had a p-value greater than 0.05. This meant the series did not influence the system, so it could potentially be removed from consideration when building our models. Below are some of the notable results of the Granger causality tests

- “days_pm2.5” had a p-value of $0.1576 > 0.05$ for Granger cause of “temperature_min”
- “precipitation” had a p-value of $0.1701 > 0.05$ for Granger cause of “temperature_max”
- “hazardous_days” had a p-value of $0.0744 > 0.05$ for Granger cause of “days_co”

Thus, we decided to remove the “temperature_min”, “temperature_max”, “precipitation”, and “days_co” features from the dataset, and we retained the “days_pm2.5” and “hazardous_days” variables because they showed a high correlation with the honey bee colony numbers. After removing the series that failed the tests, we reran the Granger causality tests again, and the remaining series all passed.

Next, we performed the Johansen cointegration test on all the series that passed the Granger causality tests. When two or more time series are cointegrated, it means they have a long run statistically significant relationship. So, the cointegration tests helped establish the presense of statistically significant connections between two or more time series. The table below contains the results of the tests

Variable Name	Test Statistic	Confidence Interval (95%)	Significance
colonies_inventory	1245.13	143.6691	True
unhealthy_days	898.63	111.7797	True
very_unhealthy_days	634.63	83.9383	True
hazardous_days	408.58	60.0627	True
days_no2	206.57	40.1749	True
days_ozone	95.59	24.2761	True
days_pm2.5	12.78	12.3212	True
days_pm10	0.44	4.1296	False

All series demonstrated significant relationships over time except “days_pm10,” so we decided to drop the feature. Therefore, we were left with the following features: “unhealthy_days”, “very_unhealthy_days”, “hazardous_days”, “days_no2”, “days_ozone”, and “days_pm2.5”.

4.2 Vector Autoregression (VAR) - Dataset A

With Dataset A, one approach was to use the aggregated time series of the remaining features to help forecast the annual honey bee colony numbers of the aggregated time series ($y_t = y_{1,t} + \dots + y_{39,t}$). Because we were dealing with multivariate time series, we decided to experiment with the Vector Autoregression (VAR) model, which can be used when two or more time series influence each other. In the VAR model, each variable was modeled as a linear combination of past values of itself and the past values of other variables in the system. Since we had several time series that influenced each other, VAR was modeled as a system of equations with one equation per variable or time series. For example, the second order VAR(2) model for two variables includes up to two lags for each variable, and it can be written as

$$\begin{aligned} Y_{1,t} &= \alpha_1 + \beta_{11,1}Y_{1,t-1} + \beta_{12,1}Y_{2,t-1} + \beta_{11,2}Y_{1,t-2} + \beta_{12,2}Y_{2,t-2} + \epsilon_{1,t} \\ Y_{2,t} &= \alpha_2 + \beta_{21,1}Y_{1,t-1} + \beta_{22,1}Y_{2,t-1} + \beta_{21,2}Y_{1,t-2} + \beta_{22,2}Y_{2,t-2} + \epsilon_{2,t} \end{aligned}$$

To select the order (P) of the VAR model, we iteratively fit several orders of the VAR model, and we picked the order that produced a model with the lowest Akaike Information Criterion (AIC) score.

Continuing on, the VAR model required the separate time series to be stationary, meaning the statistical properties of each time series (e.g., mean, variance, etc.) remained constant overtime. So, we performed the Augmented Dickey-Fuller (ADF) test, one of the most commonly used statistical tests when analyzing the stationarity of a time series, on each of the remaining aggregated time series. The first iteration of the ADF test showed that most of the time series were not stationary. However, all the time series were stationary after first-differencing. So, after applying a first-difference adjustment to the remaining aggregated time series, we were able to build the VAR model.

4.3 Long Short-Term Memory (LSTM) - Dataset A

The long short-term memory (LSTM) recurrent neural network was another methodology that we tested using Dataset A. Though, instead of aggregating the data across the 39 states and producing a single aggregate forecast, like we did with VAR, we utilized the separate data of all 39 states to produce individual forecasts of annual honey bee colony numbers for each of them. We chose LSTM because it is effective in learning the dependencies in a sequence of predictions. By using its memory cells to store relevant information, and gates that address the continuous read, write, and reset of the memory cells, it can handle both linear and nonlinear time series. It also offered the capability to optimize internally using the “Adam” optimization algorithm for each of its parameters and equations. “The Adam optimization algorithm is an extension to stochastic gradient descent that has recently seen broader adoption for deep learning applications...” [24].

In our implementation of LSTM, we decided to use the rectified linear unit (ReLU) as our activation function, one of the most popular ones for neural networks. ReLU was chosen over Sigmoid and Tanh because those activation functions can suffer saturation over time and impact the learning ability of the neural network. The ReLU function was applied to the output gate when the cell state (long-term memory) and the hidden state (short-term memory) were combined to generate the output. ReLU transformed the weighted input to the output gate in order to learn both linear and non-linear patterns in the data. Then, it generated a positive and approximately linear output using $\text{relu}(x) = \max(0, x)$. Furthermore, we employed the “Dropout” probabilistic regularization method to reduce overfitting in the neural network. During each training sample, Dropout was fed a specific probability of 20%, and it deactivated different inputs and recurrent connections to the LSTM units when the weights were updated. This allowed the LSTM units to generalize since they were less sensitive to specific weights.

To obtain the best fit model, mean squared error was chosen as the primary loss function. Also, the same features that were selected to build the VAR model were used in our LSTM approach (i.e., “unhealthy_days”,

“very_unhealthy_days”, “hazardous_days”, “days_no2”, “days_ozone”, and “days_pm2.5”). With the use of its architecture, LSTM was able to account for the trends and seasonalities that may have been present in the different time series for the feature and response variables. The retained data was structured in matrix format, 10 training epochs were used to create and fit the model, and we used a lag of 3 years as the rolling window to perform cross-validation. Then, LSTM determined the model with the minimum mean squared error, and it saved this “best” version with a call back function. Finally, we treated each of the 39 states individually to obtain the best LSTM model for each one.

4.4 Exploratory Data Analysis - Dataset B

In our project proposal, one of the methodologies we proposed was the pooled ordinary least squares (OLS) regression model, which was directly tied to a linear panel data model

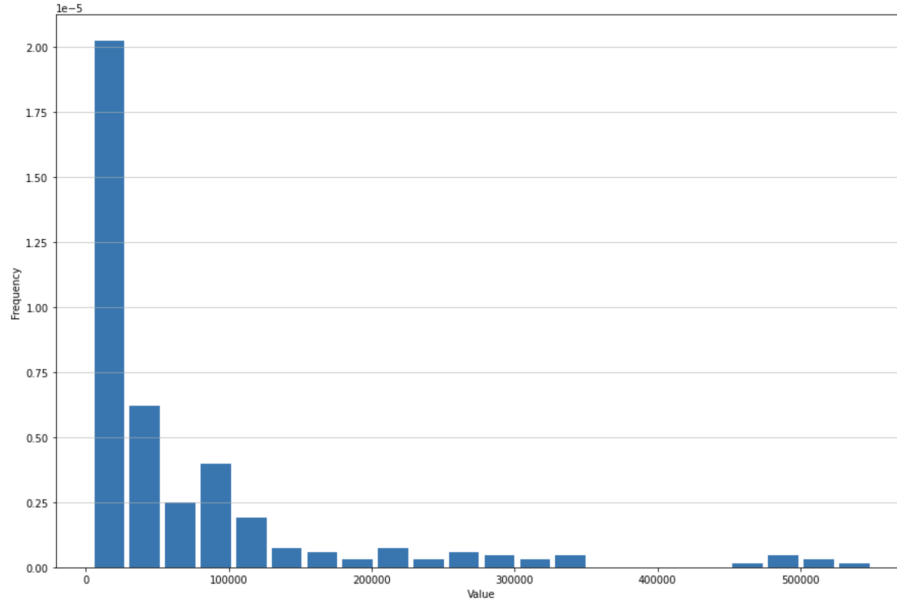
$$y_t = x_t \beta + \varepsilon_t$$

where $t = 1, \dots, T$. With the panel data setup described in the Section 3.2, we were able to write the model for an observational unit i in matrix form

$$\mathbf{y}^i = \mathbf{X}^i \beta + \varepsilon^i$$

where β was the $n \times 1$ population parameter that we want to estimate (same for all t), and ε^i was the $T \times 1$ vector of error terms.

However, with linear a model, we needed to make normality assumptions about the response data we were trying to estimate. If the response data was not close to fitting the normal distribution, then the model could give biased results. So, we first examined the histogram of the response variable:



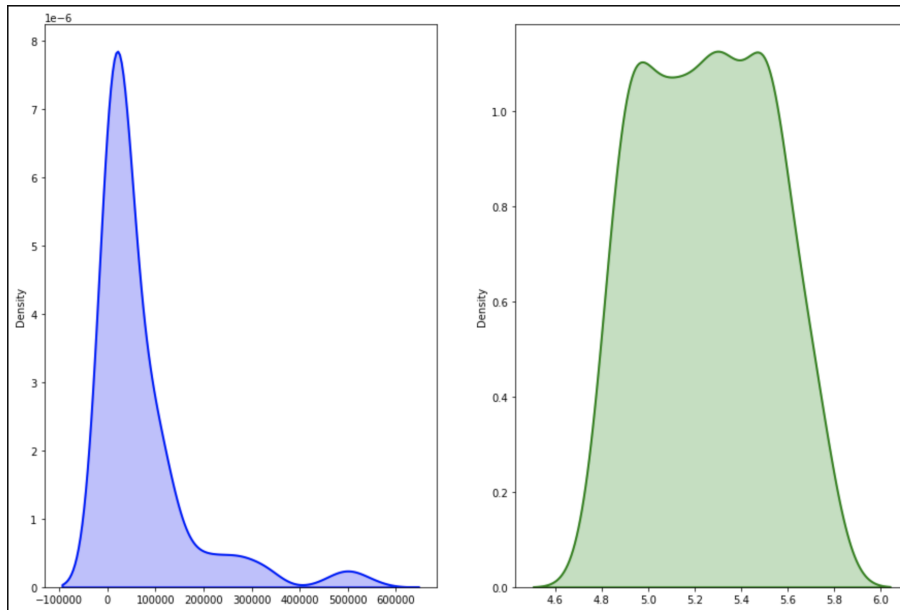
Clearly, the response variable was not normally distributed.

So, we tried to alleviate this problem by trying a Box-Cox transformation. A Box-Cox transformation is a logarithmic transformation that stretches out the smaller range to enlarge its variability and shrinks the larger range to reduce its variability

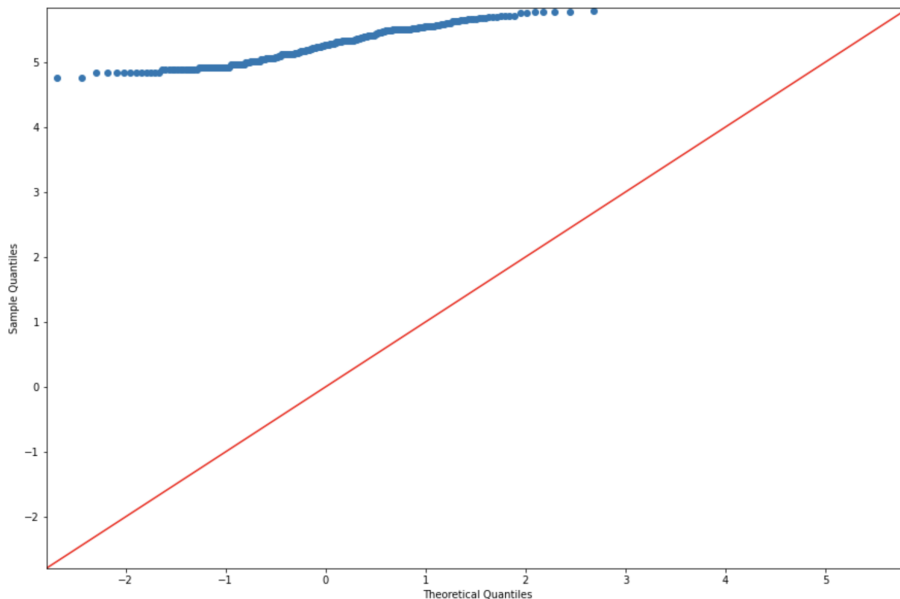
$$t(y) = \frac{(y^\lambda - 1)}{\lambda}$$

where y is the vector of responses and $t(y)$ is the transformed vector. The idea is to find the best λ such that $t(y)$ can become closer to the normal distribution.

After applying a Box-Cox transformation on the response variable, we found the optimal $\lambda^* \approx -0.1487$. Still, we needed to verify if the transformed data was approximately normally distributed:



The figure on the left is the plot of the KDE of the original data, and the figure on the right is the plot of the KDE of the transformed data. Unfortunately, the transformed data was not normally distributed, and this notion was corroborated after examining the Q-Q plot:



Therefore, we decided to change one of the original methodologies we outlined in our project proposal. Since the response variable was not normally distributed, even after applying a Box-Cox transformation, we felt it was best not to experiment with linear panel data regression models, like pooled OLS. Instead, we thought a non-linear panel data regression model would be more appropriate, and this is further discussed in the next subsection.

4.5 Mixed-Effects Random Forest (MERF) - Dataset B

Dataset B was entirely comprised of longitudinal clusters since there were multiple measures of each state over time. So, each of the 39 states represented a separate cluster. With this clustered data, we wanted to learn about the common effects across all states, but also account for the unique characteristics of each one. It was difficult to generate a separate model for each state because any given model did not have enough training data. If we created one global model to learn across all states, then it would not be able to learn anything distinctive about particular states. Also, one-hot encoding the states was an option, but it has been documented that high cardinality categorical variables as features can lead to poor model performance.

Thus, a mixed-effects model was chosen for Dataset B. “In a mixed-effect model, each cluster gets a random effect that is learned but drawn from a prior that is itself learned from the data” [25]. Mixed-effects linear modeling is a classic technique, but as was shown in the previous section, it was not a good idea to fit a linear model on Dataset B. Instead, we decided to combine random forests with linear random effects, which is known as a mixed-effects random forest (MERF) method. MERF is an extension of the random forest model to the case of clustered data, and it is implemented using a standard random forest algorithm withing the framework of the expectation-maximization (EM) algorithm [26].

The MERF method assumes a generative model of the form:

$$y = f(X) + b_i Z + \epsilon$$

where y is the response variable, $f(\cdot)$ is a general non-linear function, X contains the p fixed-effect features, Z contains the q random-effect features, ϵ is independent and identically distributed (i.i.d.) noise distributed as $N(0, \sigma_\epsilon^2)$, i is the cluster index (there are 39 clusters in Dataset B), and b_i contains the random effect coefficients. Note there are different coefficients per cluster i , but they are assumed to be drawn from the same distribution $N(0, \sigma_b)$ [25]. Generally, $f(\cdot)$ can be any non-linear regression model, but we focused on utilizing a random forest, just as in the paper by Hajjem, Bellavance, & Larocque [26].

The MERF Python Package used the EM algorithm to fit four parameters: $f(\cdot)$, b_i for all 39 clusters, σ_b , and σ_ϵ . The generalized log-likelihood (GLL) was used as a measure of training loss, and it was computed at every iteration. As the GLL decreased, the fit improved, and once it flattened, the fit converged. Just like a classic random forest model, the MERF was fairly robust to overfitting. Finally, the fit MERF was used to predict data in the 39 clusters the model saw in training:

$$\hat{y} = f(X) + b_i Z$$

where the prediction included the random effect correction.

4.5.1 Feature Selection

Now, the air quality features retained for the VAR and LSTM models (i.e., “unhealthy_days”, “very_unhealthy_days”, “hazardous_days”, “days_no2”, “days_ozone”, and “days_pm2.5”) were also considered for the MERF model, while all the other air quality and temperature features were disregarded. Yet, we still needed to identify the most relevant features among the set of variables that usually have an affect on honey bee mortality or colony numbers. Luckily, random forests are often used for feature selection for other machine learning models. The reason is because the tree-based strategies used by random forests naturally ranks features by how well they improve the purity of the node. Each tree of the random forest can calculate the importance of a feature according to its ability to increase the pureness of the leaves. “The higher the increment in leaves purity, the higher the importance of the feature. This is done for each tree, then is averaged among all the trees and, finally, normalized to 1” [27]. So, we first trained a standard random forest model

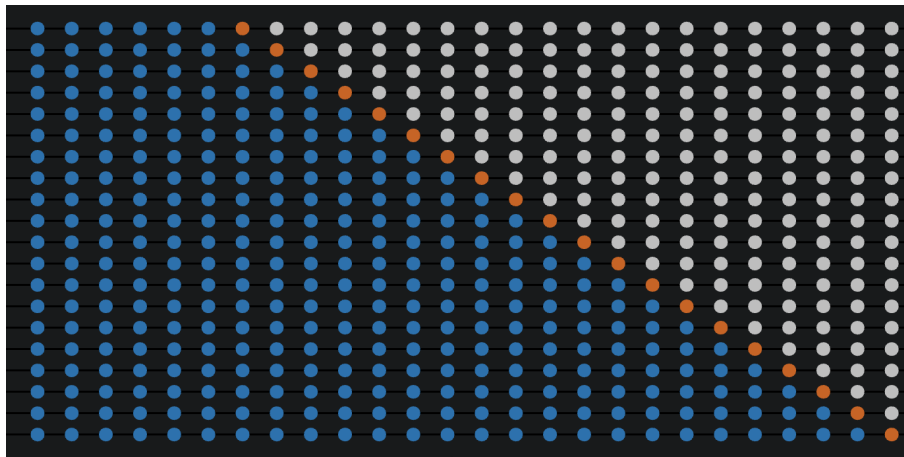
using the large set of features. Then, we identified the most important features, created a new dataset with the limited set of features, and trained the MERF model on the new dataset. Finally, because all features for each state changed over time, they were used as both fixed-effects (X matrix) and random-effects (Z matrix) in the construction of the MERF model.

5 Evaluations and Final Results

5.1 Overview

To assess the performances of the different models we tested, it was important to evaluate their accuracy using genuine forecasts. We were unable to randomly assign data into training and test sets because it made no sense to use future data to forecast values in the past. We needed to consider the temporal dependencies between the observations in our two datasets. So, particular care was taken in splitting the data to prevent data leakage. With both datasets, our data was organized chronologically, and then we picked 2016 as the point in time to split our data. All data prior to and including 2016 was always used to fit our models, while data occurring after 2016 was withheld for validation.

More specifically, data from 2017 to 2021 was used for time series cross-validation (TSCV) in which cross-validation was performed on a rolling basis. We primarily implemented the TSCV procedure outlined by Hyndman and Athanasopoulos [28] involving one-step ahead forecasts. The series of validation sets consisted of all observations from a single year, while the corresponding training sets consisted of all observations that occurred in the past. As a result, no future observations were used in the construction of forecasts. The diagram below illustrates the series of training and validation sets, where the blue dots represent the observations in the training sets and the orange dots represent the observations in the validation sets.



For a given model, the first iteration of TSCV used training data all the way up to 2016 to fit the model, a set of forecasted data points for 2017 was produced, and the accuracy of the forecasted data was measured against the validation set of true 2017 data. In subsequent iterations, the previous iteration's validation set was added to the training data to fit the model, a set of forecasted data points one year ahead was produced, and the accuracy of the forecasted data was measured against the validation set of true data one year ahead.

Thus, our TSCV process consisted of 5 separate folds since data from every individual year between 2017 to 2021 was used for validation. The overall forecast accuracy of a specific model was computed by averaging over the results generated from the 5 validation sets. It is important to note that the “one-step ahead” TSCV procedure was executed for every model we considered. This allowed us to reliably compare models that were built using either of our two datasets. Then, the model with the smallest average forecast

accuracy measures was chosen to be our best model. Finally, we trained our chosen model using the entirety of its corresponding dataset, and we estimated the true performance of this final model by evaluating its “one-step ahead” forecasts on unseen test data from 2022.

Now, when comparing the forecasting performance of different predictive models, we wanted to pick the one with the smallest average forecast error, as stated above. According to Hyndman and Athanasopoulos [28], forecast error is the difference between an observed value and its forecast, and it can be written as

$$e_{T+h} = y_{T+h} - \hat{y}_{T+h|T}$$

where the training data is given by $\{y_1, \dots, y_T\}$ and the test data is given by $\{y_{T+1}, y_{T+2}, \dots\}$. In fact, there were several different measures that we could have used to evaluate forecast accuracy. Since the forecast errors were on the same scale as the data (i.e., the total number of honey bee colonies), it was permissible to use scale-dependent accuracy measures. The two popular versions we used were root mean squared error (RMSE)

$$\text{RMSE} = \sqrt{\text{mean}(e_t^2)}$$

and mean absolute error (MAE)

$$\text{MAE} = \text{mean}(|e_t|)$$

“A forecast method that minimizes the MAE will lead to forecasts of the median, while minimizing RMSE will lead to forecasts of the mean” [28]. We also needed an accuracy measure to compare the forecast performances of models built using Dataset A versus Dataset B. So, mean absolute percentage error (MAPE) was another accuracy measure we employed:

$$\text{MAPE} = \text{mean}(|100e_t/y_t|)$$

This is because “Percentage errors have the advantage of being unit-free, and so are frequently used to compare forecast performances between datasets” [28]. Therefore, the ideal “best” model should minimize the average RMSE, MAE, and MAPE measures computed using our 5-fold TSCV procedure.

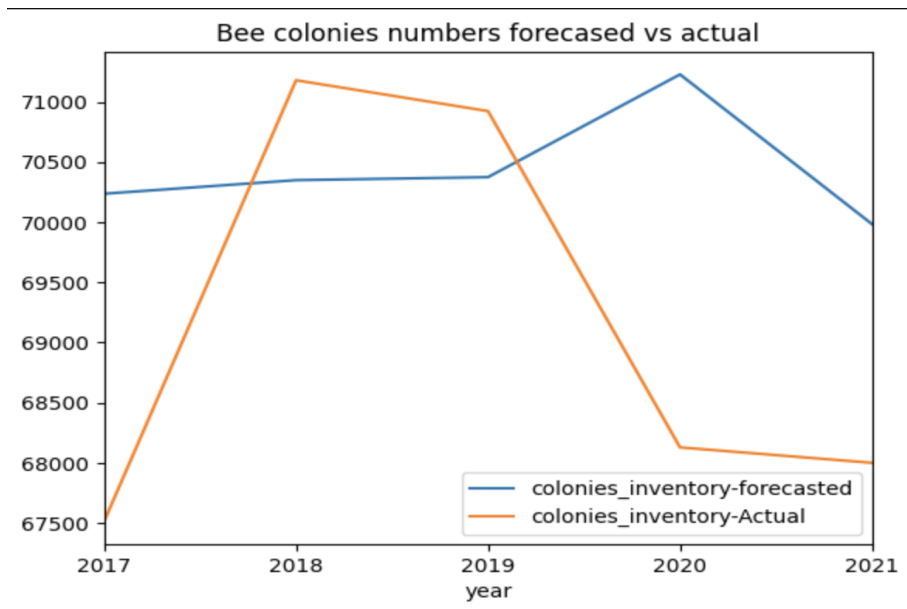
5.2 Modeling Results and Comparisons

5.2.1 Vector Autoregression (VAR) - Dataset A

First, we selected the order (P) of the VAR model. The table below contains the results of the iterative analysis. We used a lag of $P = 2$ to build the VAR model since it achieved the minimum AIC, FPE, and HQIC measures.

	AIC	BIC	FPE	HQIC
0	10.51	10.85*	3.681e+04	10.61
1	11.12	13.81	7.712e+04	11.92
2	7.419*	12.46	4494*	8.971*

Next, we performed our 5-fold TSCV procedure on the VAR(2) model. As mentioned in section 4.2, we fitted the VAR(2) model using the set of aggregated time series generated from Dataset A. From 2017 to 2021, we compared the forecasted values produced by the VAR(2) model ($\hat{y}_t$) against the actual honey bee colony numbers of the aggregated time series ($y_t = y_{1,t} + \dots + y_{39,t}$):

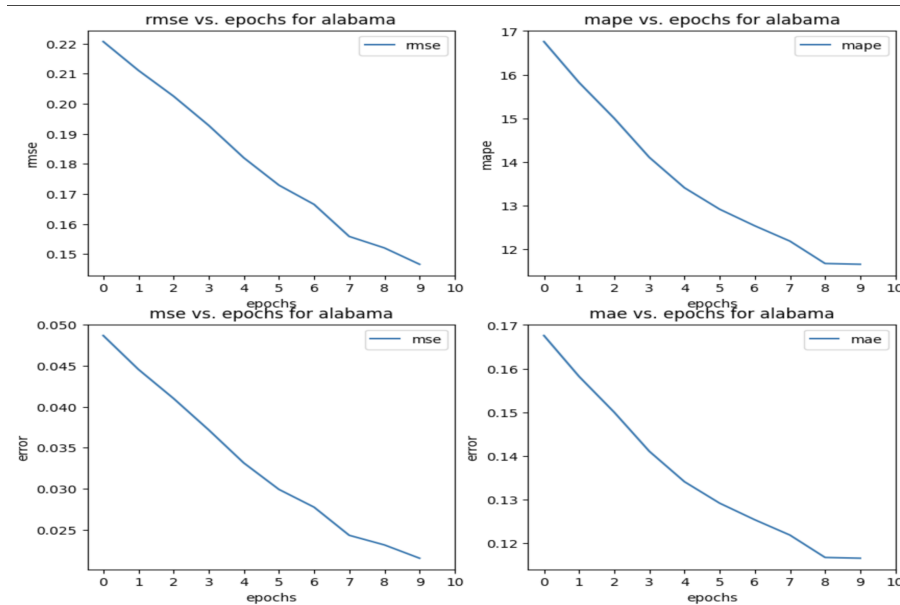


Finally, we calculated the RMSE, MAE, and MAPE of the VAR(2) model for each validation year, and we computed the average of each accuracy measure across the 5 years.

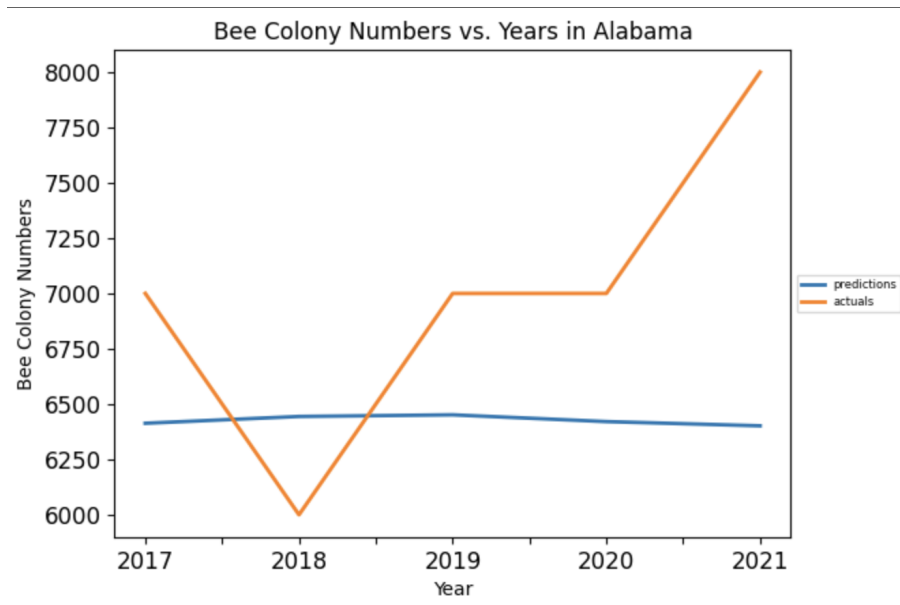
Aggregated Level Accuracy Measures			
	RMSE	MAE	MAPE
2017	2723.5532	2723.5532	4.034127
2018	830.2840	830.2840	1.166465
2019	548.8404	548.8404	0.7739
2020	3099.7425	3099.7425	4.5499
2021	1983.9329	1983.9329	2.9175
Avg.	1837.2706	1837.2706	2.6884

5.2.2 Long Short-Term Memory (LSTM) - Dataset A

As mentioned in Section 4.3, we created an LSTM model for all 39 states. So, we were able to plot the performance metrics as a function of the number of training epochs for each state. Here is an example of the output for one state:



We were also able to compare the forecasted values with the actual values for each state:

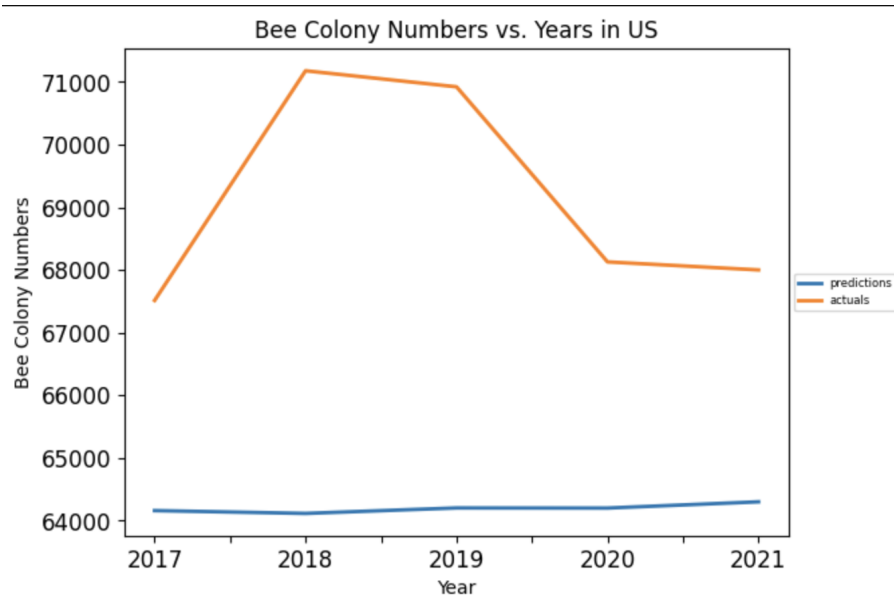


For a given validation year during our TSCV procedure, we compared the forecasted values with the actual values for all states and calculated the RMSE, MAE, and MAPE, accordingly. Then, we computed the average of each accuracy measure across the 5 years.

State Level Accuracy Measures			
	RMSE	MAE	MAPE
2017	10321.8832	5093.3003	0.1077
2018	16478.3582	7318.5767	0.1030
2019	14296.6625	6911.1679	0.1232
2020	11226.5056	6624.0724	0.1189
2021	9007.5936	5441.5851	0.1330
Avg.	12266.2006	6277.74048	0.11716

Moreover, we needed to compare the results of the LSTM approach with the results of the VAR(2)

model. So, for each validation year, we summed the forecasts of all 39 states to produce a singular aggregated forecast. Then, we compared these aggregated forecasts against the actual honey bee colony numbers of the aggregated time series ($y_t = y_{1,t} + \dots + y_{39,t}$):

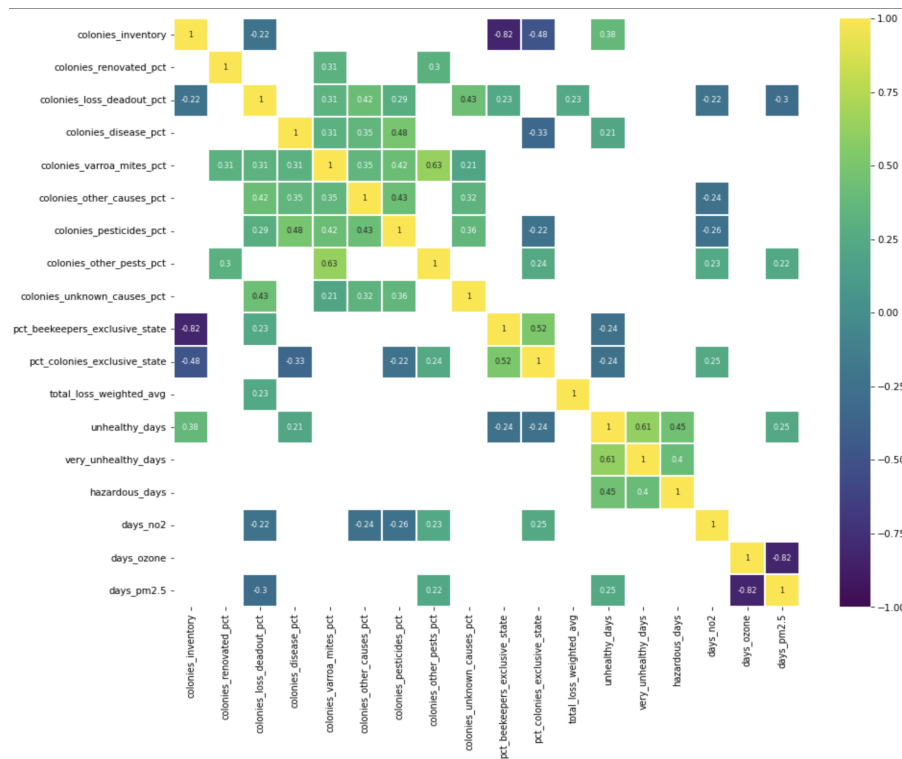


Finally, we calculated the RMSE, MAE, and MAPE for each validation year, and we computed the average of each accuracy measure across the 5 years.

	Aggregated Level Accuracy Measures		
	RMSE	MAE	MAPE
2017	3354.9885	3354.9885	0.0497
2018	7066.3700	7066.3700	0.0993
2019	6722.8660	6722.8660	0.0948
2020	3929.7637	3929.7637	0.0577
2021	3701.9219	3701.9219	0.0544
Avg.	4955.18202	4955.18202	0.07118

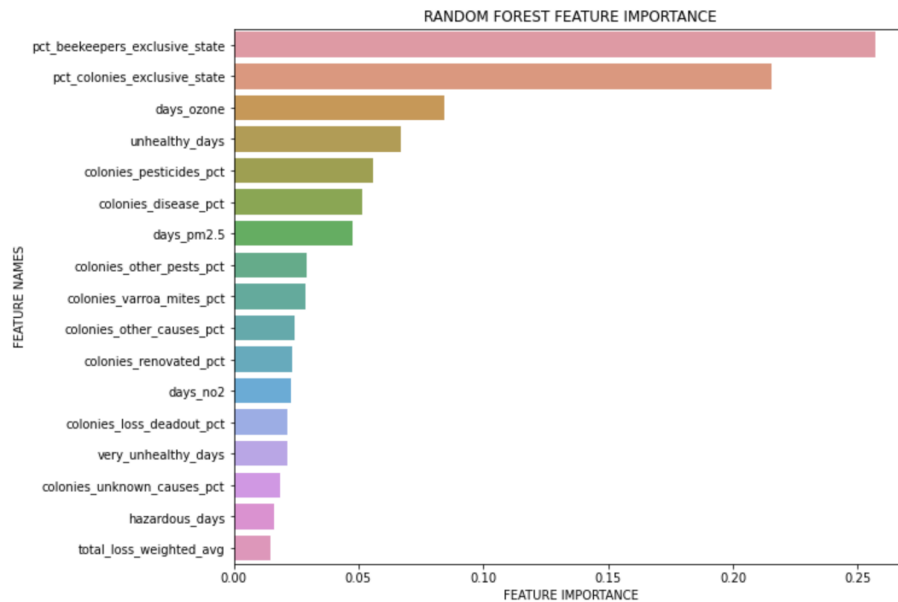
5.2.3 Mixed-Effects Random Forest - Dataset B

First, we needed to perform feature selection. Before running the random forest model, we created a correlation coefficient matrix to see how the features were related to the response variable and one another. Any correlation coefficient greater than or equal to 0.2 was displayed in the heat map below:



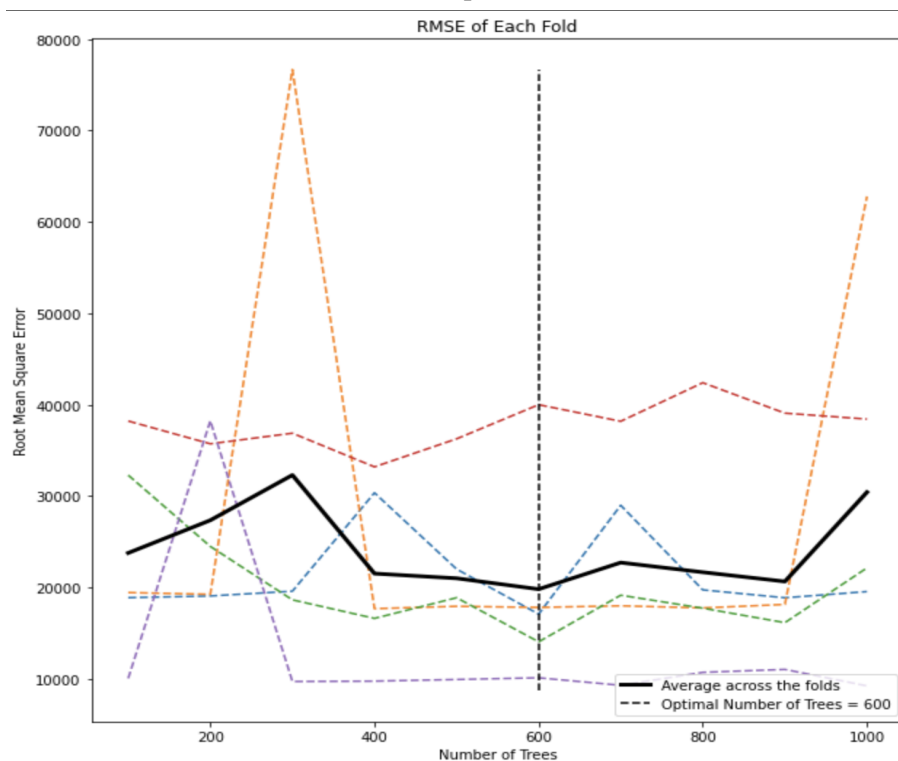
It seems as though “colonies_loss_deadout_pct”, “pct_beekeepers_exclusive_state”, “pct_colonies_exclusive_state”, and “unhealthy_days” were the features that had the strongest relationship with the response variable. With “colonies_loss_deadout_pct”, the percentage of colonies affected by “deadout loss” was negatively correlated with honey bee colony numbers. So, this feature and the response variable move in opposite directions, which was expected. Unfortunately, “pct_beekeepers_exclusive_state”, “pct_colonies_exclusive_state”, and “unhealthy_days” all had directions that were opposite of what was expected. To elaborate, we would expect that the percentage of beekeepers and colonies exclusive to a state would be positively correlated with honey bee colony numbers, while “unhealthy” air quality days would be negatively correlated with honey bee colony numbers. However, all the correlations are based on the entirety of Dataset B, meaning all the states have been lumped together. Ideally, the directional issues with these three features should be sorted out by the MERF model when we account for the differences between states.

Next, we utilized the “feature importance” attribute of the random forest algorithm. We trained a standard random forest model on years 2015 and 2016, to prevent data leakage, using the large set of features and the default parameter values. Then, we created a bar chart to display the importance of the various features:



Based on the two sets of results that were displayed, we decided to retain all of the features with an importance rating of 0.05 and above, along with “colonies_loss_deadout_pct” based on its correlation with the response variable.

Now, the EM algorithm used to fit the MERF model often took quite a long time to run, even with a “stop early” threshold and maximum limit on the number of iterations. Also, due to the TSCV procedure, the MERF model needed to be fit 5 separate times. Consequently, it was difficult to perform hyper-parameter tuning because of these computing constraints. Hence, we only focused on tuning the number of trees for the random forest, while the rest of the parameters were set to their default values.



After determining the optimal number of trees, we reran the TSCV procedure with this value inputted as a

set parameter in the MERF model. For a given validation year during our TSCV procedure, we compared the forecasted values with the actual values for all states and calculated the RMSE, MAE, and MAPE, accordingly. Then, we computed the average of each accuracy measure across the 5 years.

State Level Accuracy Measures			
	RMSE	MAE	MAPE
2017	18761.3905	11936.5135	0.6788
2018	17746.5537	9804.6502	0.3148
2019	14208.1014	8493.3901	0.2868
2020	14163.7961	8174.7696	0.2362
2021	16390.5067	11765.5029	0.3447
Avg.	16254.0697	10034.9652	0.3723

Finally, we compared the aggregated forecasts against the actual honey bee colony numbers of the aggregated time series ($y_t = y_{1,t} + \dots + y_{39,t}$):

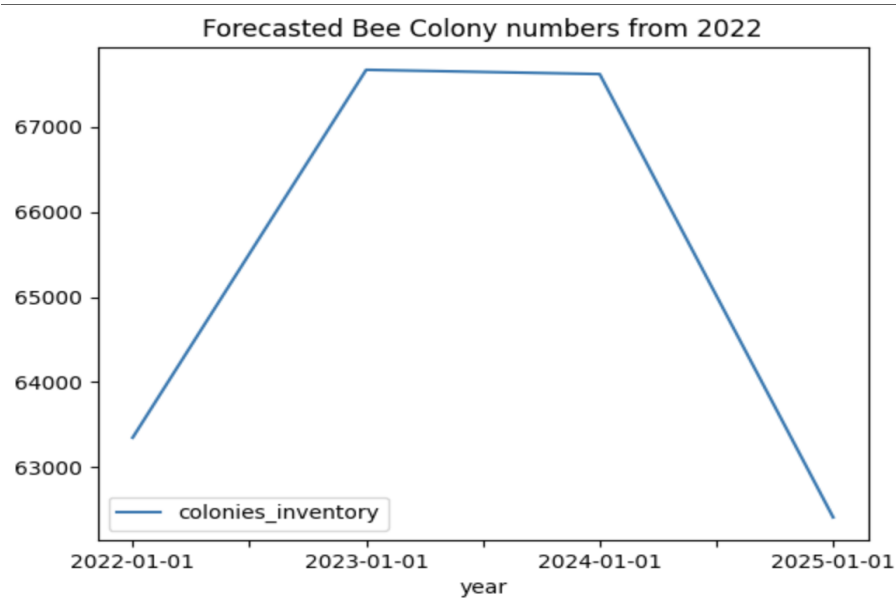
Aggregated Level Accuracy Measures			
	RMSE	MAE	MAPE
2017	133604.6305	133604.6305	0.0507
2018	157897.6551	157897.6551	0.0569
2019	40970.4556	40970.4556	0.0148
2020	93146.6446	93146.6446	0.0351
2021	40908.5355	40908.5355	0.0154
Avg.	93305.58426	93305.58426	0.03458

5.3 Estimated True Performance of Final Model

Based on the results above, we chose the VAR(2) model to be the best one out of the three. It had the lowest average RMSE value, and its forecasted values actually changed in a notable way from year to year, unlike the LSTM model. Thus, we fit the VAR(2) model using data from all years, and we estimated its true performance by evaluating its “one-step ahead” forecast on unseen test data from 2022.

	Actual	Forecast	RMSE
2022	72673.076923	63348.063277	9325.013646

Finally, we created a forecast of annual honey bee colony numbers over the next four years



6 Conclusion

For the aggregated data from Dataset A, the VAR(2) model had an RMSE of 1837.2706, and the LSTM model had an RMSE of 4955.18202, which meant the VAR(2) model performed better at the aggregated level. As a result, we proposed the VAR(2) model for predicting the annual honey bee colony numbers for the U.S. using the variables “unhealthy_days”, “very_unhealthy_days”, “hazardous_days”, “days_no2”, “days_ozone”, and “days_pm2.5”. From the explanatory data analysis, the most important series for forecasting the colony inventories were the “pm2.5” and “hazardous_days” features. Remember, PM2.5 is a measure of the tiny particles in the air that reduce visibility when levels are elevated, and higher measures of PM2.5 does have a notable impact on the honey bee population. The “hazardous_days” feature represented the number of days when the air had elevated pollutants beyond the normal threshold. Therefore, we can conclude that pollution does have a substantial impact on honey bee colonies, specifically in the United States. Based on the VAR(2) model forecasts, the honey bee colony population will rise to approximately 70,000 in 2023, but it will decrease to approximately 62,000 in 2025. Therefore, we recommend businesses and government authorities to take the appropriate measures if they wish to improve the honey bee population in the future.

With Dataset B, models such as VAR and LSTM were not suitable because the dataset had very few years in the time series in order for these models to learn successfully. While we did try to fit the mixed-effects random forest on Dataset B, it did not produce good forecasts, which was confirmed by its high RMSE values. Hence, decided to discard Dataset B from our final considerations.

7 Work Breakdown

All three team members contributed equally to this project. We all collected and processed data, built and tested models, and wrote various portions of the proposal and final report.

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